

Text & speech data

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The Golden Rule of Data Science



“Garbage in – Garbage out”

→ In many ways choosing the right data and preparing it appropriately is the most important step

Obtaining the right data

- What type of data will be needed?
- Is the data sensitive?
- How much data will be needed?
 - Very difficult to predict
 - Depends on task difficulty and model
 - Adding more data often easiest way to improve model
- Are there commonly used benchmarking datasets?

What types of text and speech data
do we have in medicine and life
sciences?

Write down what you use or could use for your research!

Language data in medicine and life sciences

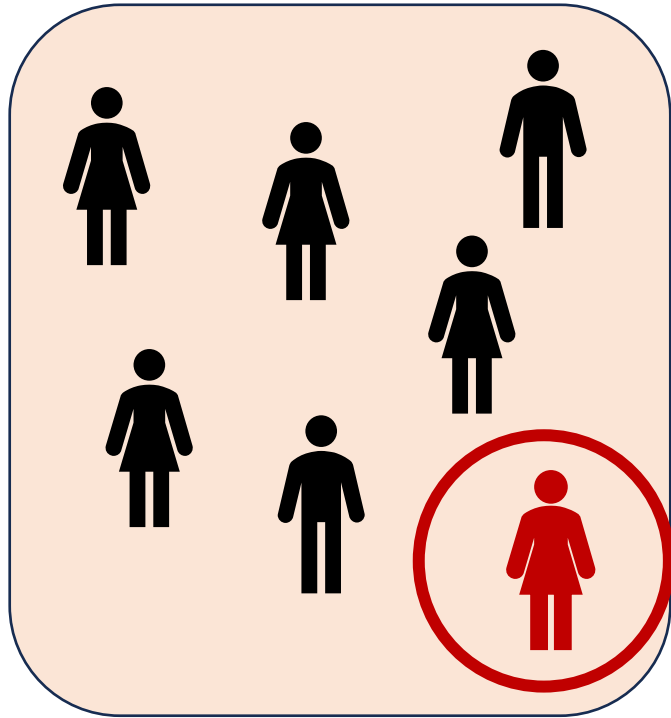
- Patient-doctor conversations
- Emergency calls
- Electronic health records
- Literature – abstracts or fulltext
- Reports
- Patents
- Social media posts
- Experiment descriptions
- Database entries

...

What is data?

What is data?

Dataset



Example



Feature
Age
Sex
Height
Weight
Diabetic

Value
35
Female
1.70 m
78 kg
Yes

Types of data

Big data $\leftrightarrow$ Small data

Single instance $\leftrightarrow$ Sequence

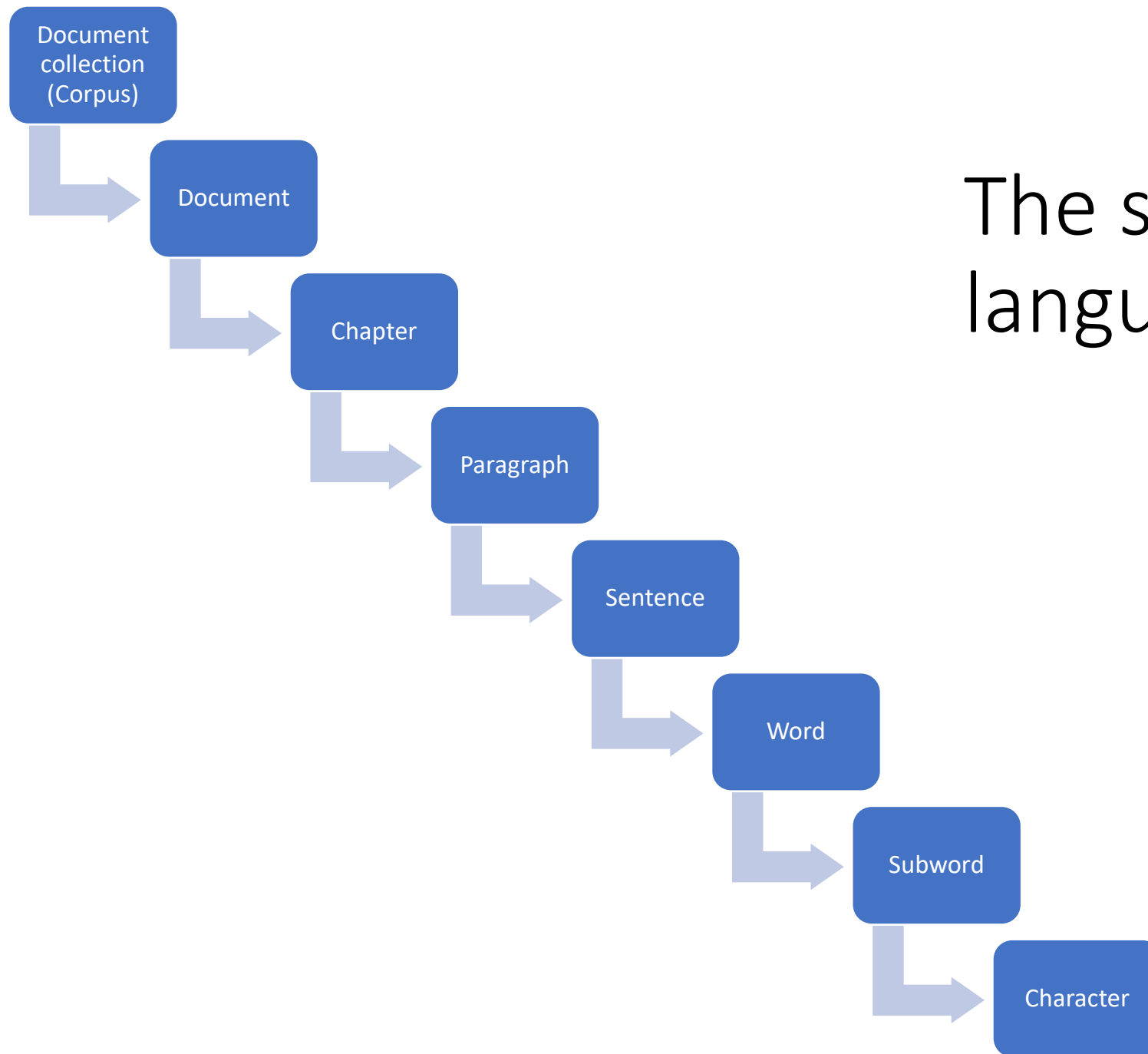
Labelled (annotated) $\leftrightarrow$ Unlabelled

Unstructured $\leftrightarrow$ Structured

Sensitive $\leftrightarrow$ Not sensitive

Raw $\leftrightarrow$ Processed

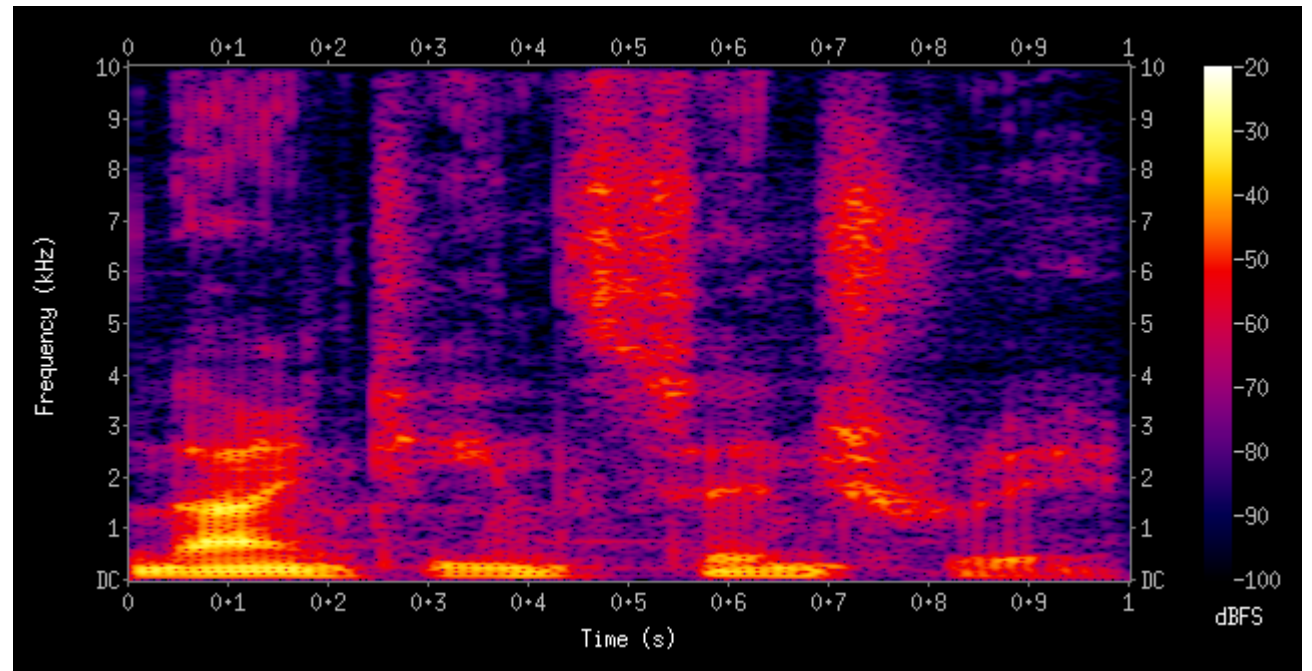
The structure of language data



Speech data contains more information than just the words

- Content: words
- Voice characteristics: speed, pitch, pauses, accent, etc.
- Background noise

Spectrogram



Finding data

- Chatbots
- Search engines
 - <https://datasetsearch.research.google.com/>
 - <https://emedia.lub.lu.se/db>
- Collections of databases and repositories
 - <https://www.re3data.org/>
 - <https://fairsharing.org/databases/>
 - <https://snd.gu.se/en/find-data/international-data>
 - <https://www.oxfordjournals.org/nar/database/a/>
 - https://en.wikipedia.org/wiki/List_of_biological_databases
 - <https://www.springernature.com/gp/authors/research-data-policy/recommended-repositories>
 - <https://www.ebi.ac.uk/>
 - <https://www.ncbi.nlm.nih.gov/>
- Data repositories
 - <https://zenodo.org/>
 - <https://figshare.com/>

Finding data

- Machine learning data collections
 - <https://www.kaggle.com/datasets>
 - <https://huggingface.co/docs/datasets/>
 - https://en.wikipedia.org/wiki/List_of_datasets_for_machine-learning_research
 - https://en.wikipedia.org/wiki/Biomedical_text_mining#Corpora
- Scientific databases
 - <https://www.gbif.org/species/> (species)
- Research infrastructures and consortia
 - <https://www.ukbiobank.ac.uk/>
 - <https://gdc.cancer.gov/>
- Governmental and non-governmental organizations
 - <https://www.who.int/data/collections>
- Data-centric journals
 - <https://www.sciencedirect.com/journal/data-in-brief>

Swedish data

- <https://www.lupop.lu.se/data-at-LU>
- [SoS](#) (Socialstyrelsen)
- [SCB](#) (Statistikmyndigheten)
- <https://rut.registerforskning.se/>
- <https://snd.gu.se/>
- <https://www.pathogens.se/>

Sensitive data must be handled with care

- Personally identifiable information (name, person number, email, phone, etc.)
- Sensitive personal data (health information, biometrical data, ethnicity, race, sexual orientation, etc.)
- Classified data
- Confidential data

→ special requirements for handling – cannot be processed on most computational infrastructures

→ **Avoid if possible!**

Artificial data can often replace/complement real data

- Fake medical records to train summarization models – avoids privacy issues
- AI-generated negative and positive statements to train classification models – avoids collecting hand-labelled data (slow and expensive)
- Synonym replacement – for data augmentation
- ...

Benchmarking datasets

- High quality
- Often published for scientific competitions
- Leaderboards for tracking progress

Examples:

Hugging Face Datasets Hub

Papers with Code – Natural Language Processing

State-of-the-art (SOTA) performance

Websites tracking SOTA

- <http://nlpprogress.com> (text)
- https://github.com/syhw/wer_are_we (speech)
- <https://paperswithcode.com/>

Community challenges (“shared tasks”)

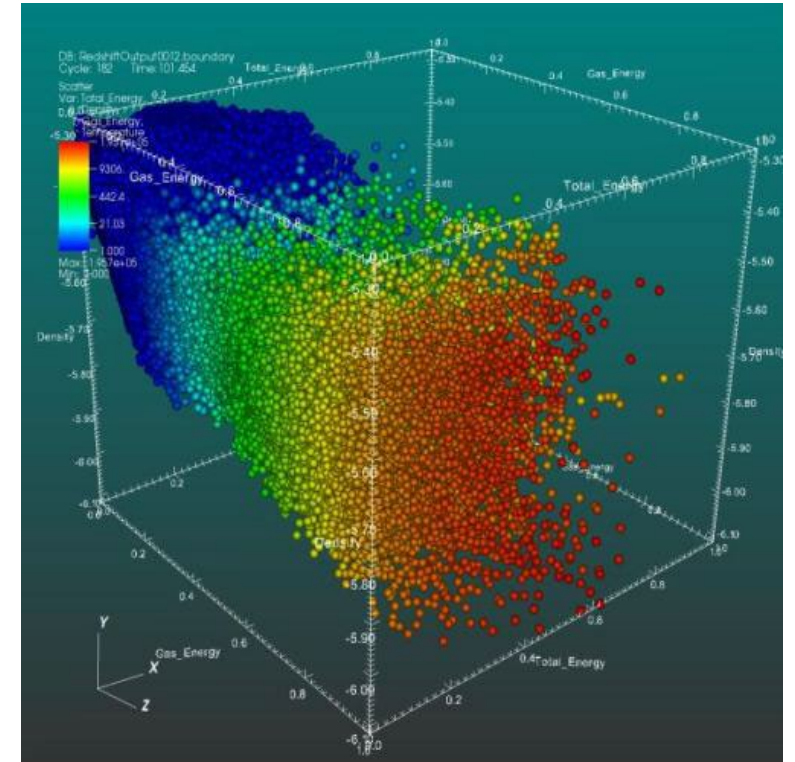
- Kaggle
- Conferences

Recent research articles

Before you prepare the data, you need to understand it!

Exploratory Data Analysis (EDA)

- **Manual inspection of raw data!**
- Visualization
- Statistical analysis



What could EDA for text/speech data include?

EDA for language data

- Language
- Number of samples
- Class distribution
- Length (word count, duration)
- Vocabulary
- Word frequency
- Voice characteristics
- Missing values
- Spelling mistakes
- ...

What problems have you encountered in text/speech data?

Data preprocessing

- Data exclusion
- Normalization
- Handling of missing values
- Dimensionality reduction
- Encoding
- Augmentation
- Splitting
- Denoising
- ...

Encoding

Data must be converted to numerical representations → Many ways to do this!

Binary encoding

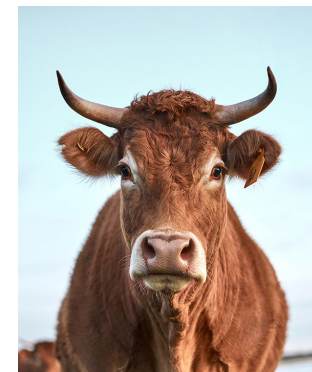
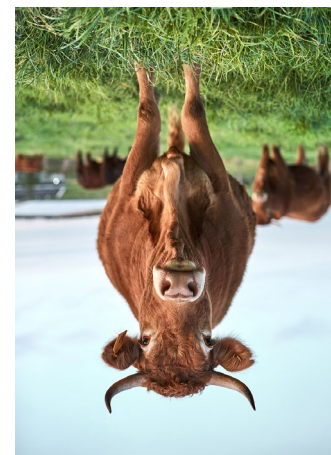
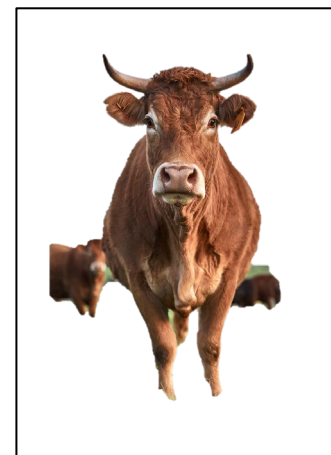
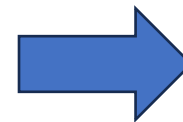
Cancer	Healthy
1	0

One-hot encoding

Sample	Diseases	Sample	D	C	S
1	diabetes, cancer	1	1	1	0
2	diabetes, stroke	2	1	0	1
3	stroke	3	0	0	1
4	cancer, stroke	4	0	1	1

Augmentation

- Introduces variation
- Increases dataset size
- Can balance data



How could you augment text or speech data?

Language data is extremely ambiguous

She killed the man with the tie.

Why is language data challenging?

- **Ambiguity**

- Homographs: We **ran** a Western blot to measure **RAN** levels.
- Homophones: to, two, too

- **Co-reference**

Cancer is one of the most common causes of death. **This disease...**

- **Synonymous expressions**

- This caused **cell death**./This led to **cellular demise**./This **killed the cells**./The **viability was greatly reduced**./The **cells were eradicated**.

- **Abbreviations**

Summary

- Medicine and life sciences has a broad variety of text and speech data
- Garbage in – garbage out!!!
- EDA should inform data preprocessing
- Language data can be broken down into substructures
- Speech data is more than word content